

David Shumway

Portfolio & Example Projects | August 2026

LLM-Assisted Crowdsourcing Marketplace Codebase Analysis | Independent Research Project, 2026

- Developed an LLM-based microtask classification pipeline to categorize 500+ JavaScript automation blocks across a 14-class domain taxonomy, engineering structured prompts with sliding-window conversation history and a domain-specific architectural cheat sheet to guide a local Llama3 model toward reliable JSON output.
- Built a git history redaction scanner to prepare an 8-year, 3,700-commit codebase for academic publication, combining regex-based detection of PII, API keys, and infrastructure fingerprints across all commit diffs with incremental save/resume, observability, folder-level filtering, and automated generation of a git filter-repo replacements file.
- Designed a OWL ontology capturing marketplace, worker, task, and infrastructure dynamics, enabling LLM-driven SPARQL querying and pattern discovery toward natural language query answering.

Tech: Claude, Llama 3B (local), GPU inference, Python, Jupyter, Git, Bash, JSON, embeddings, prompt engineering

Modular Insurance CRM Platform | Scalesology, 2025-2026

- Led backend development of a greenfield insurance CRM, migrating legacy workflows to a scalable Python/FastAPI architecture on AWS (250K+ LOC), including completion of roughly 8 out of 12 of the major backend system modules and design and implementation of 50-100 API endpoints
- Designed relational schema and managed PostgreSQL RDS infrastructure on AWS (RDS, EC2, S3); built core service modules supporting complex CRM workflows via SQLAlchemy and n8n.
- Collaborated cross-functionally to translate evolving client requirements into backend APIs and data models, and proactively flagged and mitigated system-level design risks.
- Leveraged LLMs throughout the development lifecycle, generating PRs (GitHub Copilot), system design, documentation, unit tests, and FastAPI/SQLAlchemy code, and iterated on Copilot PR feedback until optimal solutions were achieved within time constraints, with targeted human tweaks to ensure full integration.
- Maintained a sustained contribution cadence of 150-250 commits per month over 7 months, reflecting deep ownership of the codebase and continuous delivery of features.
- Designed and implemented a dynamic policy coverage calculator, an Excel-like formula engine for insurance valuations. Built an initial AST-based parser to evaluate complex conditional formulas, then transitioned to a fully transparent implementation with 75-125 explicit Python functions per client requirements. The engine automatically recalculated/propagated dependent coverage values when any referenced field changed.

Tech: Python, FastAPI, SQLAlchemy, PostgreSQL, AWS (RDS, ECS, EKS, S3, CloudWatch, Route 53, IAM, Secrets Manager, Cognito), n8n, Git, GitHub Copilot, React

React Joyride Tour + Components for University Recruiting App | Scalesology, 2024-2025

- Added React Joyride tour functionality to a custom university recruiting app (roughly 10 guided tour steps), working alongside a lead React developer and backend engineer (SQLAlchemy, PostgreSQL on RDS).
- Held sole responsibility for tour development, collaborating with PM/SA to design and implement client-desired functionality.
- Used D3.js alongside ECharts to implement a custom ECharts parliament chart relating the ratio of students retained versus left.

Tech: React, React Joyride, JavaScript, SQLAlchemy, PostgreSQL, AWS RDS, ECharts, D3.js

YAML Unit Testing Framework Extension | Scalesology, 2024-2025

- Added new YAML unit tests and modified the custom Python unit test runner to enable execution against additional database schemas (prod, sandbox, dev, etc.) from the command line.
- Ensured consistent formatting and readability across existing and new YAML tests for long-term maintainability.

Tech: Python, YAML, PostgreSQL

Oil Recycling Work Order Processing Migration | Scalesology, 2024-2025

- Transitioned ~50% of legacy SQL Server stored procedures to Python for work order processing, adding readability, maintainability, and structured logging to the system.
- Maintained SQL Server-to-Snowflake migration scripts and diagnosed warnings and errors.
- Took over PowerBI administration and reporting after client's analyst departed, performing both administrative duties such as enabling access for new users, and new PowerBI development such as addressing bugfixes and adding new

report functionality.

- Analyzed Snowflake data warehouse usage and reviewed DAX queries for Oracle migration planning.
- Diagnosed Snowflake security and access issues, including differentiating human vs. programmatic accounts, enforcing two-factor authentication for human users by generating and sending 2FA enrollment emails, and performing a full accounting of all human accounts on Snowflake while ensuring all Python Snowflake libraries were up-to-date.

Tech: Python, SQL Server, Snowflake, PowerBI, DAX, SQL

Tableau Reports for Financial Company | Scalesology, 2025

- Created and evolved Tableau reports tracking financial performance data for companies across two separate reports, with multiple visualizations, pages, and tabs per report.
- Built a Python ETL script to process messy spreadsheet data (copied from an external reporting tool) into clean Tableau-ready format, transforming block-style report data into standardized rows and columns.
- Data was loaded via Excel spreadsheets with multiple tabs containing company performance metrics.

Tech: Tableau, Excel, Python

Grocery Store PowerBI Proof of Concept | Scalesology, 2025

- Transitioned a large JavaScript + SQL Server report into a PowerBI proof of concept with multiple pages, tables, charts, and a custom data matrix visualization (compiled via terminal).
- Loaded SQL Server data (30-60K grocery products) into Azure SQL, then pulled into PowerBI and Fabric/OneLake for high-speed queries across millions of rows with semantic modeling and performance tuning.

Tech: PowerBI, Fabric, OneLake, Azure SQL, SQL Server, custom visualizations

Building Footprint Clustering Tool (DBSCAN → HDBSCAN) | Scalesology, 2025

- Evolved a building footprint clustering tool from DBSCAN to HDBSCAN to add confidence scores to algorithm outputs for an insurance client.
- Added Microsoft building dataset to the app using PostGIS on AWS RDS for an additional building footprint heuristic.
- Discovered and fixed redundant geocoding API calls by adding disk-based caching keyed by request parameters, reducing API costs and accelerating development iteration.

Tech: Python, HDBSCAN, PostGIS, AWS RDS, geocoding APIs

AI methods for reducing diver exposure to biological hazards | Lead Developer, 2021-5/2026

- Constructed water quality datasets by combining multiple environmental data, developed and tested machine learning, transfer learning, and domain adaptation models, including supervised, semi-supervised, and unsupervised methods, training tens of thousands of models across thousands of source sites and temporal windows.
- Leveraged GPU training (Google Colab) with PyTorch and TensorFlow for neural network experiments (MLP, LSTM, GRU, GNN), processing domain-adaptation experiments across thousands of site-specific models, and built a waterborne illness ontology to ground ML predictions.
- Built ROWIO (Recreational and Occupational Waterborne Illness Ontology), a domain-specific OWL ontology and knowledge graph integrating environmental, microbial, and health-related data through systematic reuse of established biomedical and environmental ontologies including UMLS, Infectious Disease Ontology, Environment Ontology, and the Exposure Ontology to support data integration and model interpretability. Conducted ontology embedding experiments (node2vec) to evaluate whether KG structure improved ML prediction, establishing that ontological coverage alone did not immediately confer predictive signal, but retained value as a scaffold for conceptual validation and interpretability.

Tech: Python, PyTorch, TensorFlow, Scikit-Learn, XGBoost, Pandas, GeoPandas, Dask, Google Colab (GPU), SPARQL, RDF, OWL, Protégé, Virtuoso, rdflib, OBO ROBOT, LaTeX, Overleaf, node2vec, knowledge graph embeddings, OOPS Pitfall Scanner

Visualization & Data Analytics: Pandas, matplotlib, GeoPandas, D3.js, Plotly, Google Colab + Google Drive, Google Sheets
Machine Learning & AI: time-series prediction, domain adaptation, transfer learning, supervised & unsupervised learning, Linear Regression, Random Forest, Decision Tree, SVM, MLP, CNN, XGBoost, LSTM, ADAPT, EasyAdapt, subspace alignment, CORAL, GRU, GNN, BERT, LLM

Popular APIs & Datasets: UMLS, Infectious Disease Ontology, WaterQualityData.us, EPA BEACON, OpenStreetMap API, NOAA Global Hourly, NOAA Tides & Currents, NOAA NDBC, EPA ECHO, EN4 Met Office Hadley Centre Ocean Salinity

FDIC Quarterly Banking App | Independent project, 2023

- Single-page app using Flask, JavaScript, and DataTables.js to analyze FDIC Banks API reporting data from 5K+ US

financial institutions, with mechanisms for automated retrieval, caching, and CSV download of selected views.
Tech: Flask, DataTables.js, FDIC Banks API

[MALDI-DB: A public repository for MALDI-TOF mass spectra](#) | Lead Developer, 2018-2021

- Built a community-driven public repository for preprocessing, analysis, search, and hosting of MALDI-TOF mass spectra and their small molecules, utilizing R, RPlumber, PostgreSQL, Nginx, Python, Django, D3.js, Docker, Docker Compose.
- Utilized OpenStack on MassOpenCloud to provision resources and publish a development version of the site online.

Files: <https://github.com/idbac/malddb>

Development screenshots: <https://github.com/idbac/malddb/tree/master/mdb/media>

[Zoomie: Round-robin breakout scheduling in Zoom](#) | Independent project, 2020

- Simple browser extension built for a meditation group which utilizes Zoom's web-chat client, enabling moderators to run round-robin partnered breakout room events (Firefox, Chrome), open sourced on GitHub with documentation, issue tracking, and bugfixes.

[Chicago Public Schools Enrollment App](#) | Visual Design Studio (CS526), 2018

- Single-page app built the UIC Department of Minority Affairs toward recruitment of high school students from within the City of Chicago.
- Led the development of 3 interactive visualizations and search functionality, then took the initiative to refactor the entire project into a seamless single-page application over winter break, resulting in a grade upgrade and the app being adopted as a teaching demo.

Tech: HTML, CSS, JavaScript, D3.js, Leaflet, Chicago Public Schools dataset

Files: <https://github.com/davidshumway/CPS>

[Finger Lakes Business Systems](#) | Lead Developer, 2010-2018

- Engineered multi-account, quota-aware distributed execution platform over Amazon Mechanical Turk using JavaScript, Chrome Extension API, LAMP stack, Python (Scrapy), Git, Fedora Linux; and designed factory-based browser extension architecture (~300K LOC JS, 300+ library methods, 100+ factory methods; ~50K LOC PHP, 150+ backend API endpoints), backed by a MySQL backend storing 200M+ records.
- Engineered programmatic data extraction pipelines for financial and government data sources, including FINRA profiles, EDGAR filings, Cook County Clerk and Tax portals, NYC e-Services (nycserv.nyc.gov), physician profiles (npiregistry.cms.hhs.gov), and quarterly credit union call reports.
- Implemented strategies to handle CAPTCHAs and rate limiters, maintain logged-in sessions, and retrieve structured data from complex websites.
- Integrated external data into task workflows, with automated submission and lightweight worker validation, ensuring high data quality and 99.98% approval rates across thousands of tasks while achieving hourly wages 3–6× above published benchmarks.

Tech: JavaScript, Linux, MySQL, PHP, HTML, CSS, Python, Chrome Extensions API, Firefox Add-Ons, Browser

Extensions, Amazon Mechanical Turk (AMT), Chart.js, Three.js

Files (available on request):

- AMT marketplace-related browser extensions
- AMT marketplace-related MySQL databases
- AMT marketplace-related PHP scripting
- AMT marketplace-related Apache Cordova mobile app
- AMT task-related browser extensions
- AMT task-related MySQL databases
- AMT task-related PHP scripting
- AMT task-related Python scripting (Scrapy)
- Browser extensions, SQL, and PHP scripting related to clients outside of AMT

Popular APIs used included Google Places (nearby search, text search, autocomplete, radar search), Google Maps, Yelp Businesses (v1-v3), Facebook Graph, DBpedia, Foursquare, BBB, Wikipedia, GeoNames, Twitter, Airbnb, Upcatabase.org, Smoopa.com.

[Tools for Amazon's Mechanical Turk](#) | Independent project, 2013-2018

- Developed public-facing open-source Firefox and Chrome browser extensions to improve worker productivity on the (AMT) website, utilized by up to 10,000 users per week.

Tech: Firefox / Chrome browser extension, Chrome Extension API, HTML, CSS, JavaScript

[Trumpocalypse](#) | Team project, Software Design Studio (CSC446), Fall 2017

- Politically-themed satirical text-based game in the style of Oregon Trail, built using the Python Pygame library.
- Completed roughly 350 out of 500 team commits.

[École Polytechnique Fédérale de Lausanne](#) | Summer Intern, Summer 2017

Tech: Neo4j, Cypher, Ubuntu, Node.js, PostgreSQL

Files (available upon request):

- Cypher queries for working with Kamusi's Neo4j graph database
- PostgreSQL queries for analysis of the PanLex language database

[Paul Brunton Philosophic Foundation](#) | Technology Specialist & Developer, 2010-2013

- Automated the (EPSON) scanning process, resulting in 4-8x efficiency and saving 2-4 years of manual scanning work.
- Created MySQL database full-text search form within Microsoft Word using VB.NET and Visual Studio.
- Utilized tesseract open-source OCR library to extract text from typewriter-written documents, saving the extracted text to a MySQL database, and implemented tools for 1) LibreOffice auto-complete using macros, and 2) duplicate hard-copy document checking.
- Utilized ImageMagick to automatically crop scanned white space from thousands of scanned typewritten documents, and created and used an interactive tool to validate cropped white space using thumbnails (HTML/JavaScript/CSS), providing performance gains of up to 5x over manual image cropping.

Tech: Linux, Apache, MySQL, PHP, HTML, CSS, JavaScript, VBA, VB.NET, Tesseract, Bash, Visual Studio, LibreOffice, Microsoft Word

Files (available on request):

- Epson scanning automation tool: Initially VBA macros, and later combined with VB.NET front-end
- Microsoft Word, MySQL database full-text search form (VB.NET)
- Tesseract OCR via shell scripting, MySQL database
- ImageMagick cropping and crop-review tool (LAMP, HTML, JavaScript, CSS)

[Freelance Technology Specialist](#) | Technology Specialist & Developer. 2005-2010

- Provided freelance tech services in person and on freelance sites such as Rent-A-Coder, Freelancer.com, and MTurk.com.

Tech: Linux, Apache, MySQL, PHP, HTML, CSS, JavaScript, Audio transcription